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Tadakuma

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(54) **SIMULATION MODEL AND SIMULATION METHOD**

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G06F 119/06 (2020.01)
(Continued)

(52) **U.S. Cl.**
CPC ***G06F 30/367*** (2020.01); ***H03K 17/567*** (2013.01); ***H03K 17/687*** (2013.01); ***H10D 12/481*** (2025.01); *G06F 2119/06* (2020.01)

(58) **Field of Classification Search**
CPC .. G06F 30/367; G06F 2119/06; H10D 12/481
(Continued)

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An Office Action; mailed by the China National Intellectual Property Administration of the People's Republic of China on Apr. 29, 2025, which corresponds to Chinese Patent Application No. 202210017151.2 and is related to U.S. Appl. No. 17/506,974.

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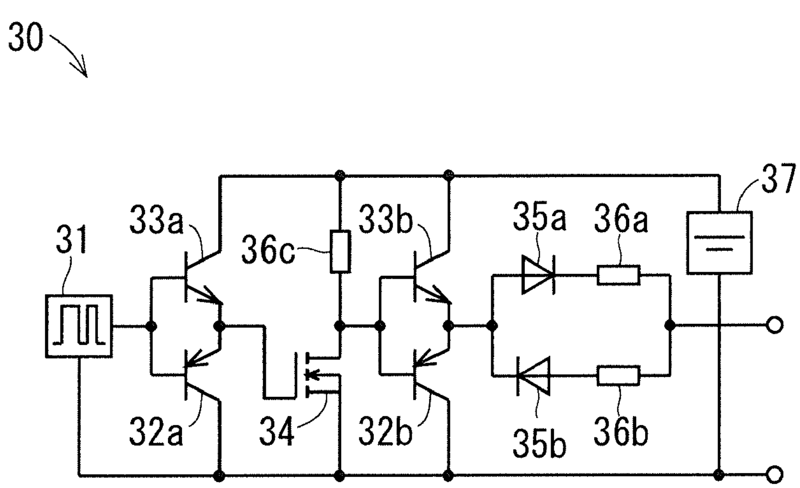
(57) **ABSTRACT**

An object of the present disclosure is to accurately simulate the operation of a CSTBT. The simulation model of a CSTBT includes a MOSFET, a diode whose cathode is connected to the drain of the MOSFET, capacitance C_{GE} connected between a source and a gate of the MOSFET, capacitance C_{CG} connected between a gate of the MOSFET and an anode of the diode, capacitance C_{CE} connected between a source of the MOSFET and the anode of the diode, capacitance C_{DG} connected between the drain and the gate of the MOSFET, and a behavioral power source V_{DG} connected in series to the capacitance C_{DG} between the drain and the gate of the MOSFET. The behavioral power source V_{DG} performs a switching operation when gate-emitter voltage V_{GE} of the CSTBT reaches a predetermined threshold value.

10 Claims, 19 Drawing Sheets



(51) **Int. Cl.**
H03K 17/567 (2006.01)
H03K 17/687 (2006.01)
H10D 12/00 (2025.01)

(58) **Field of Classification Search**
USPC 716/111
See application file for complete search history.

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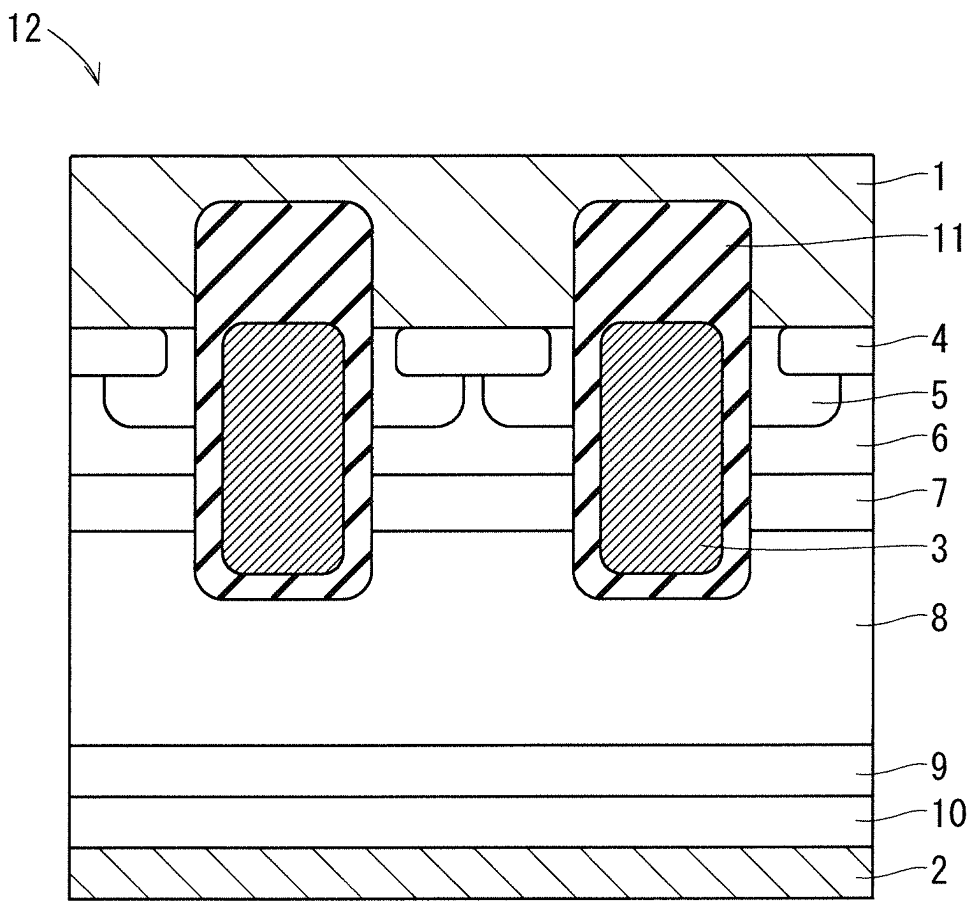
F I G. 1



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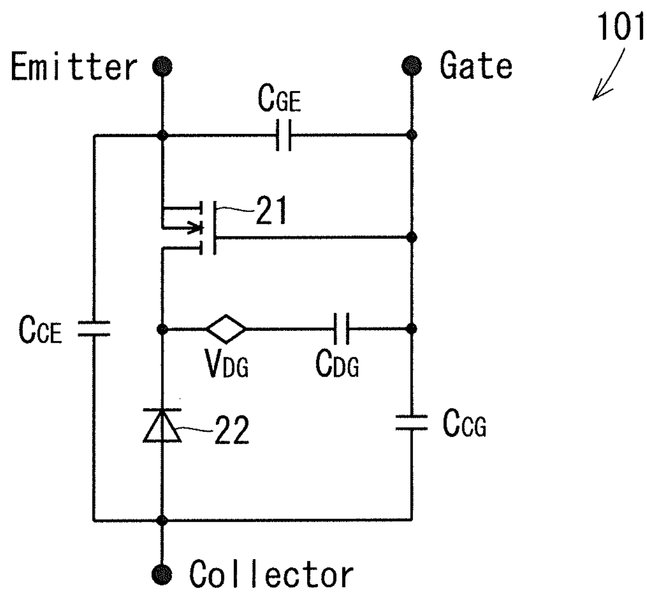

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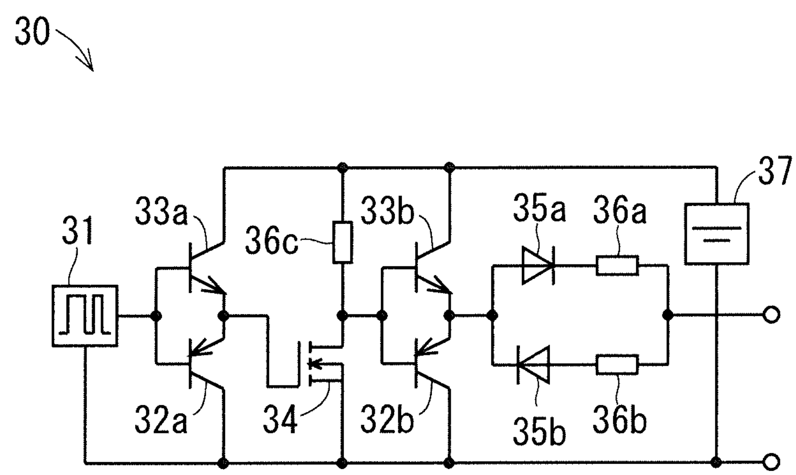

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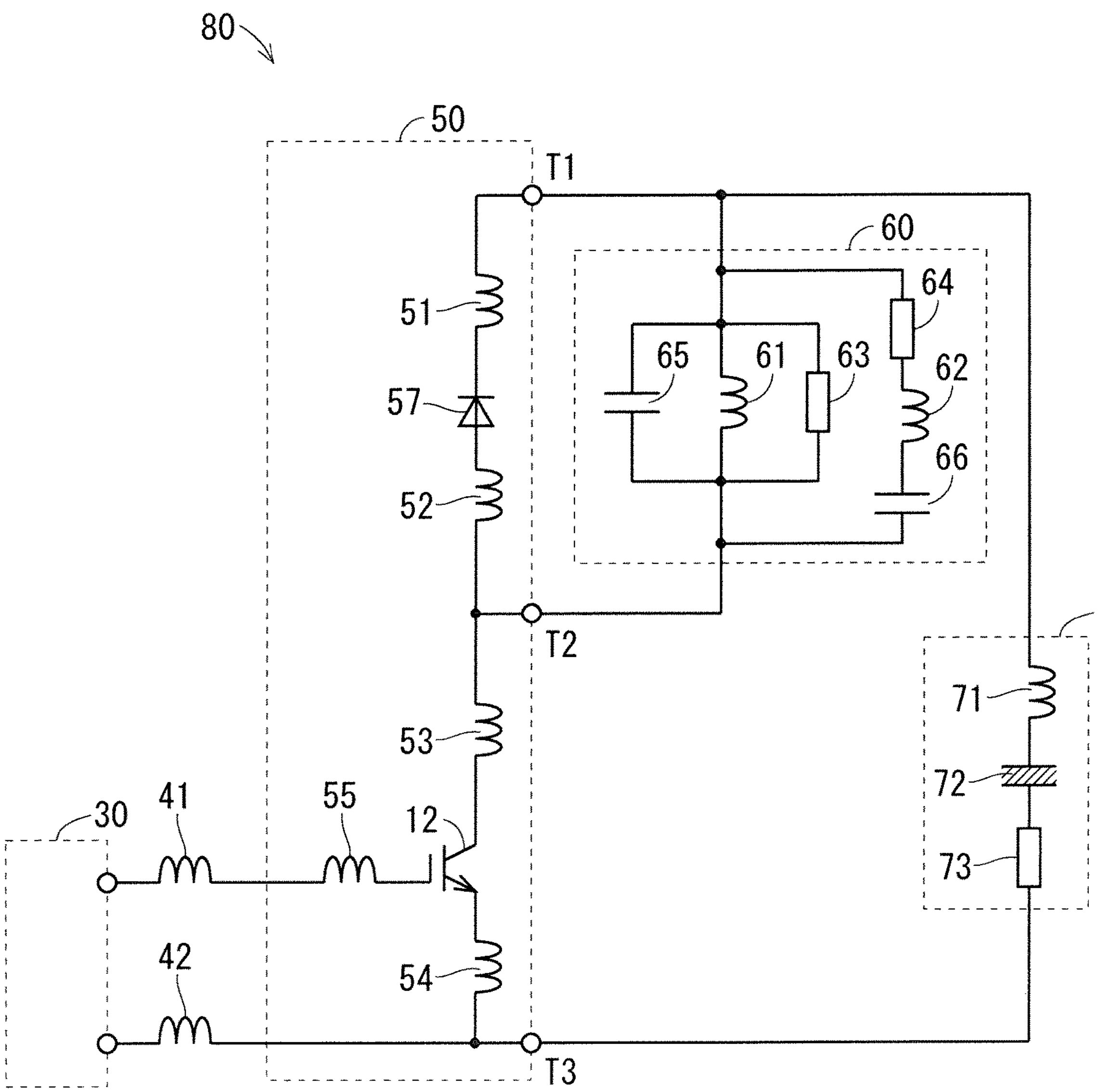
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T1
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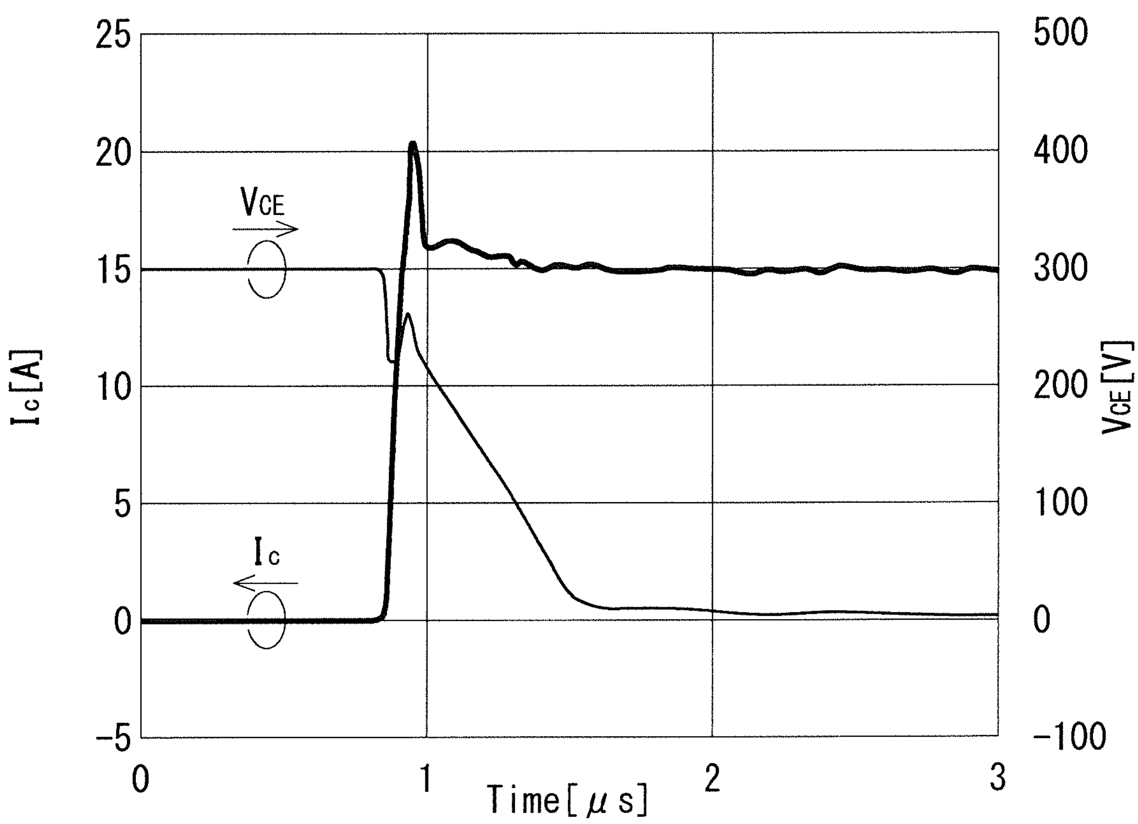
T3

FIG. 5



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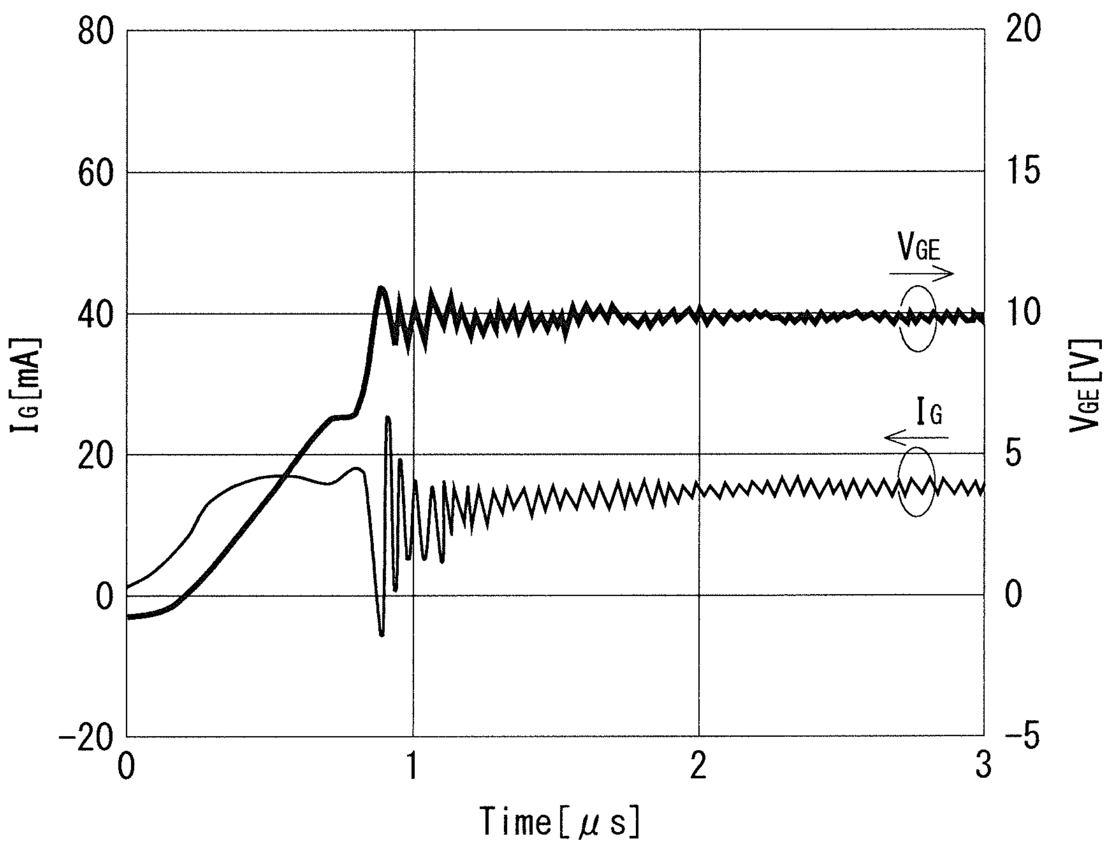
Time[μs]

FIG. 6



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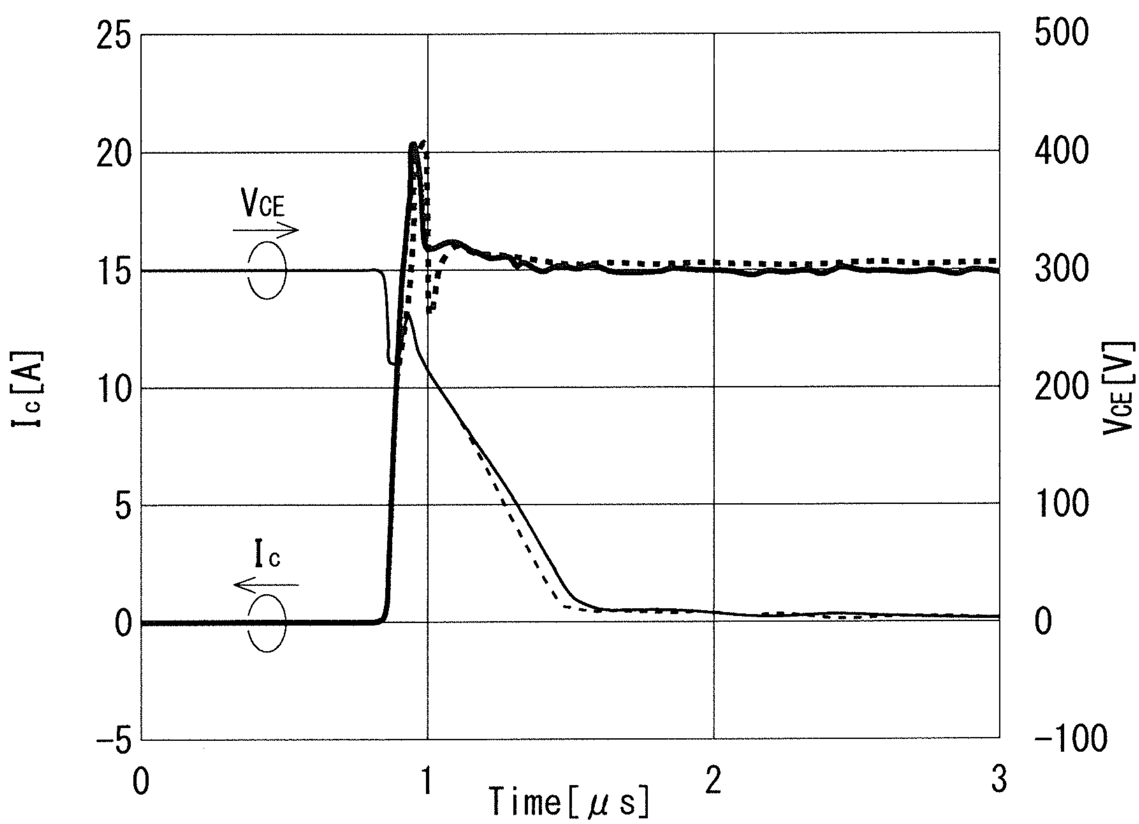

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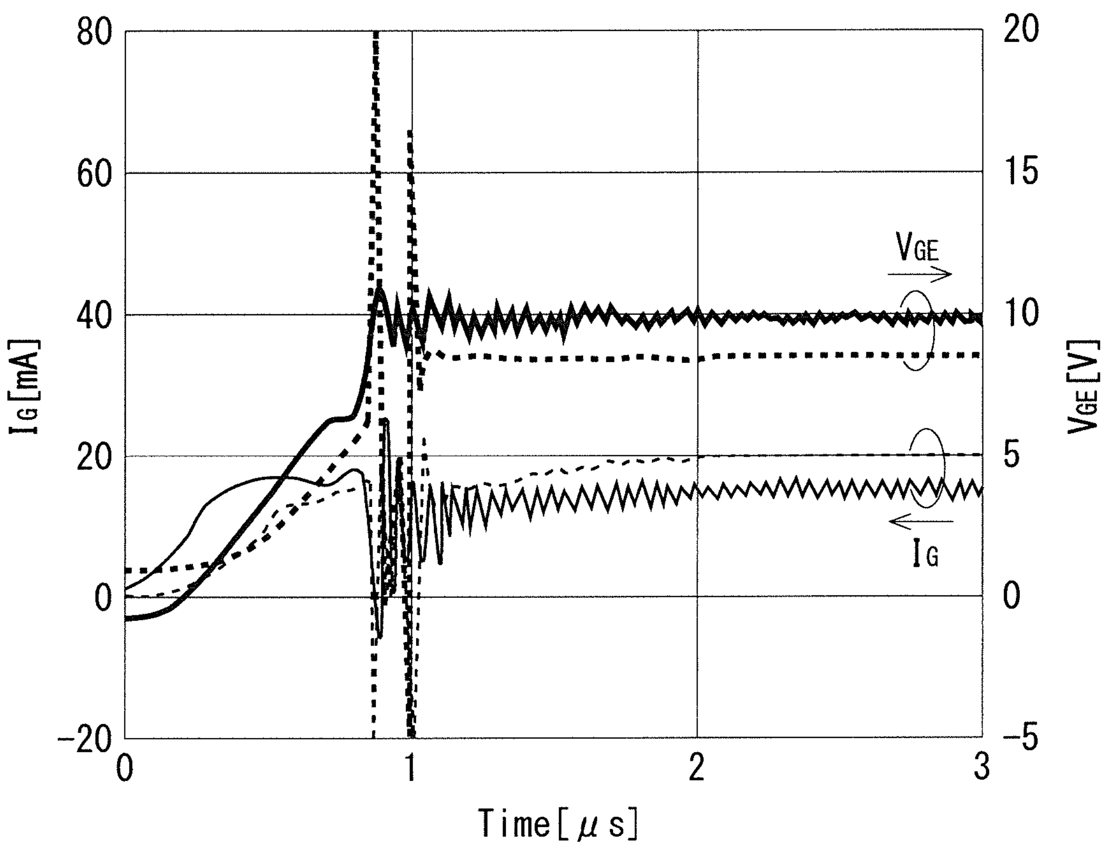
Time[μs]
Ic[A]
VCE[V]
VCE
Ic

FIG. 8



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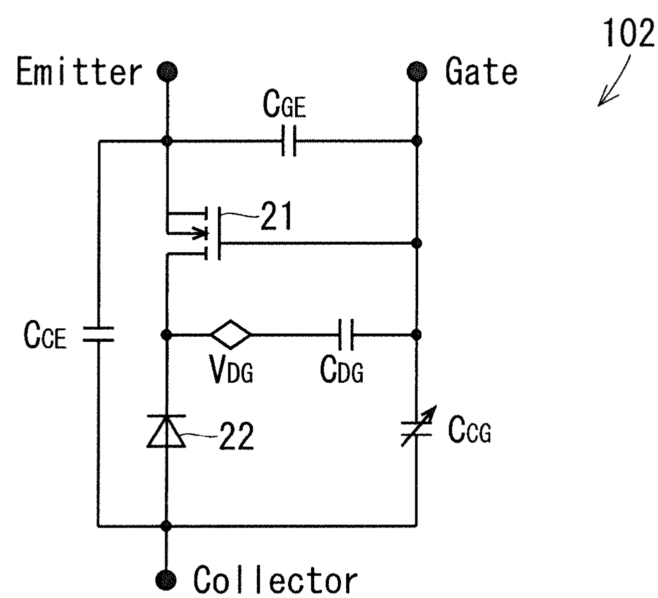


FIG. 10

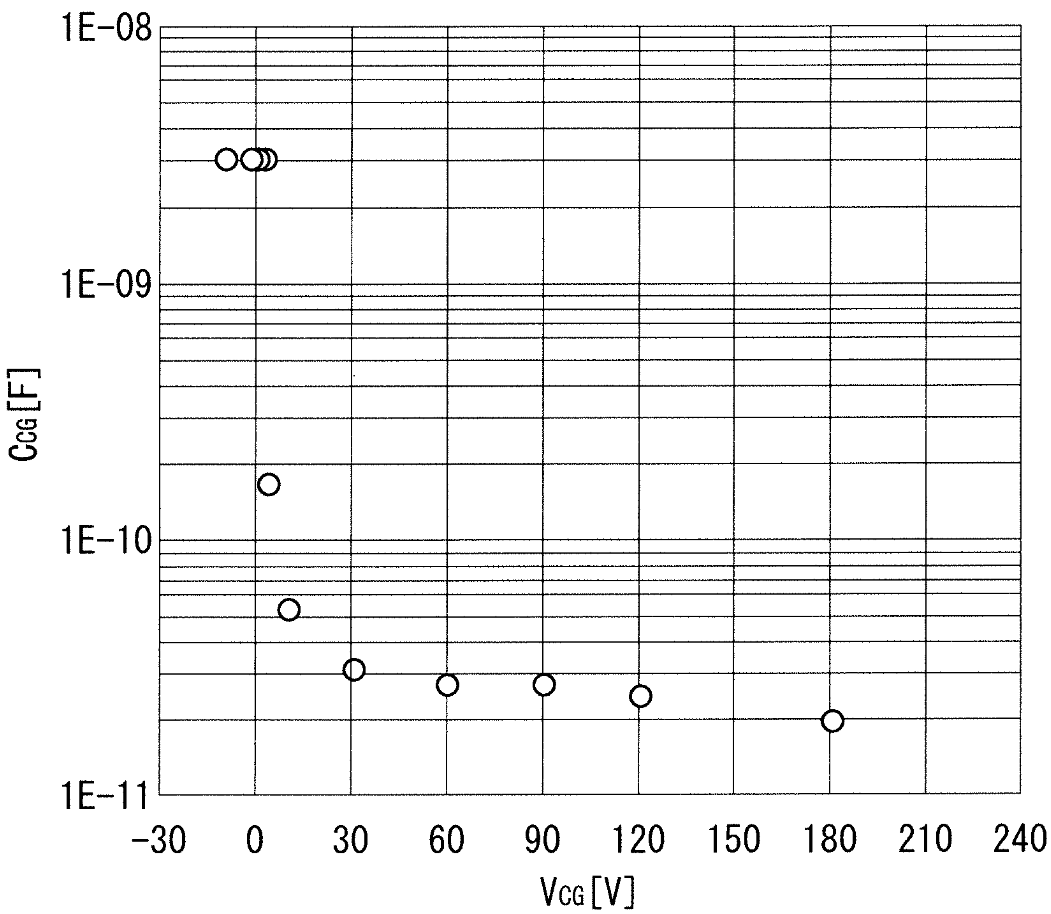


FIG. 11

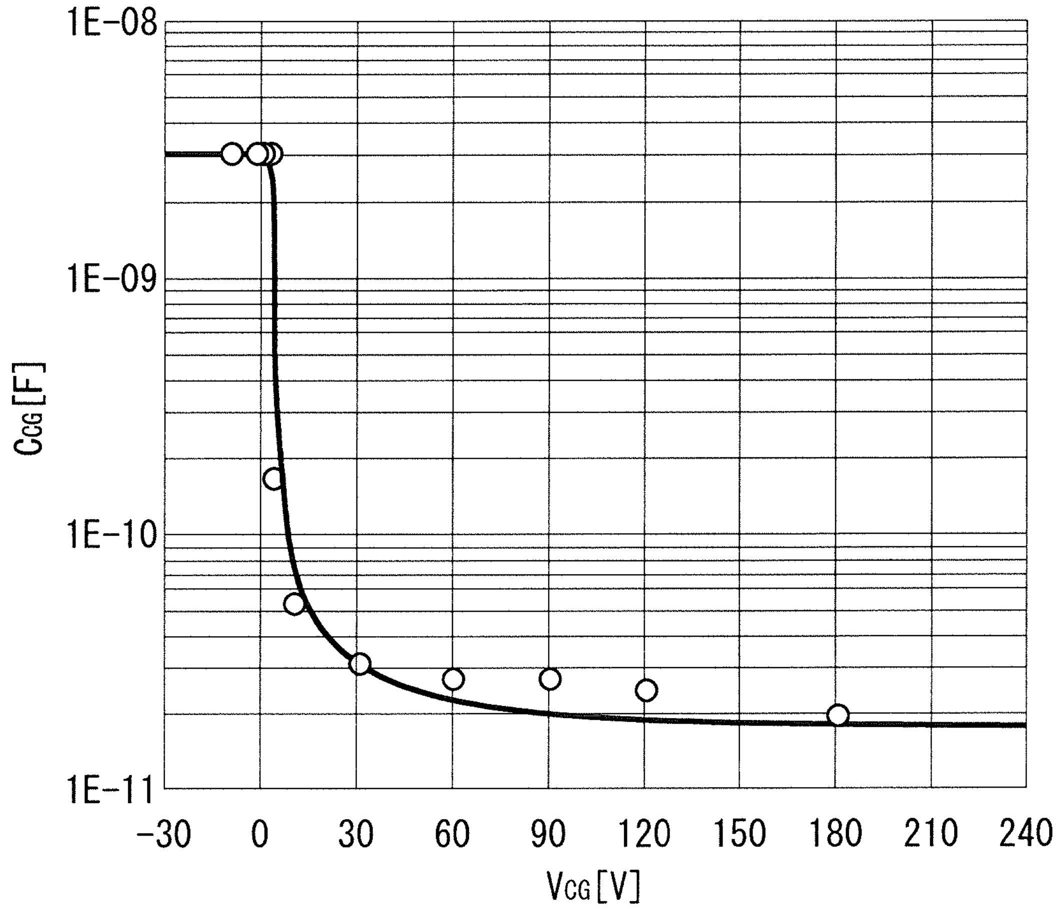
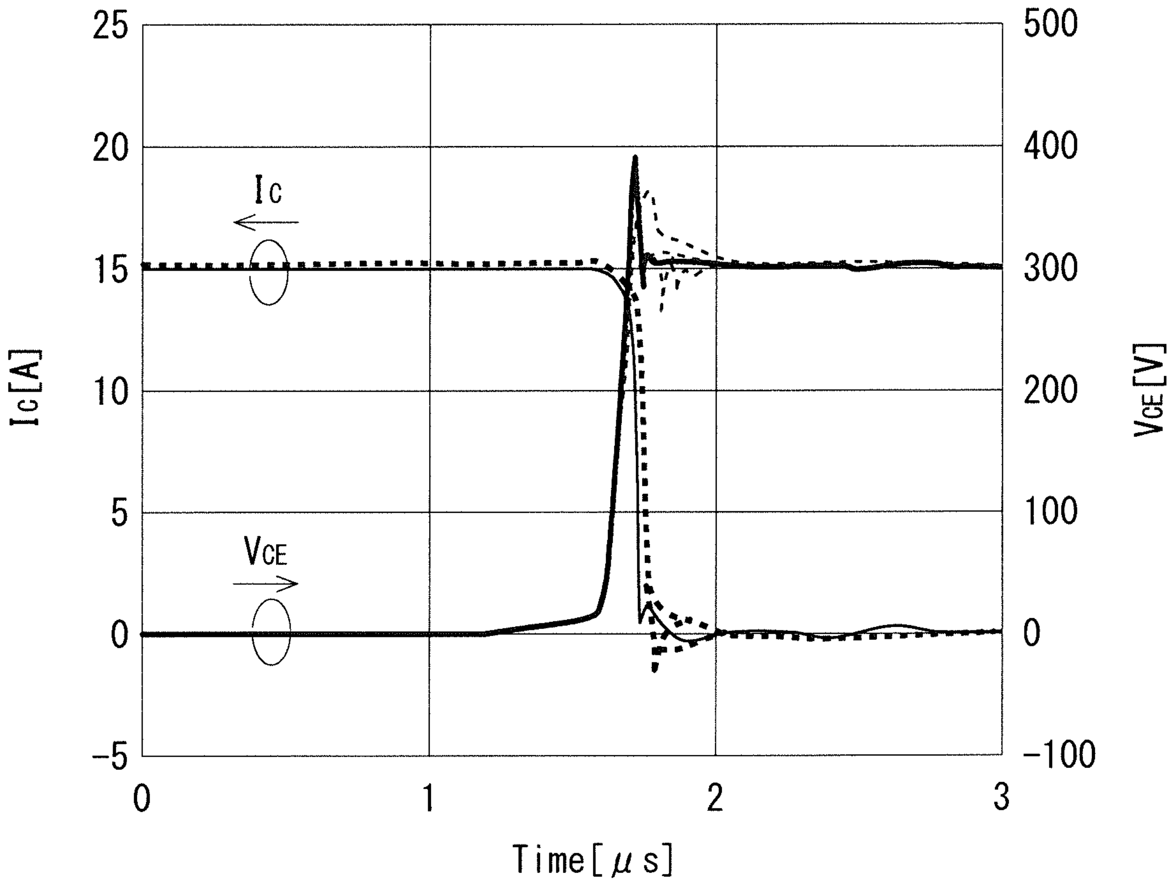


FIG. 12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Ic[A]
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Time[μs]

FIG. 13

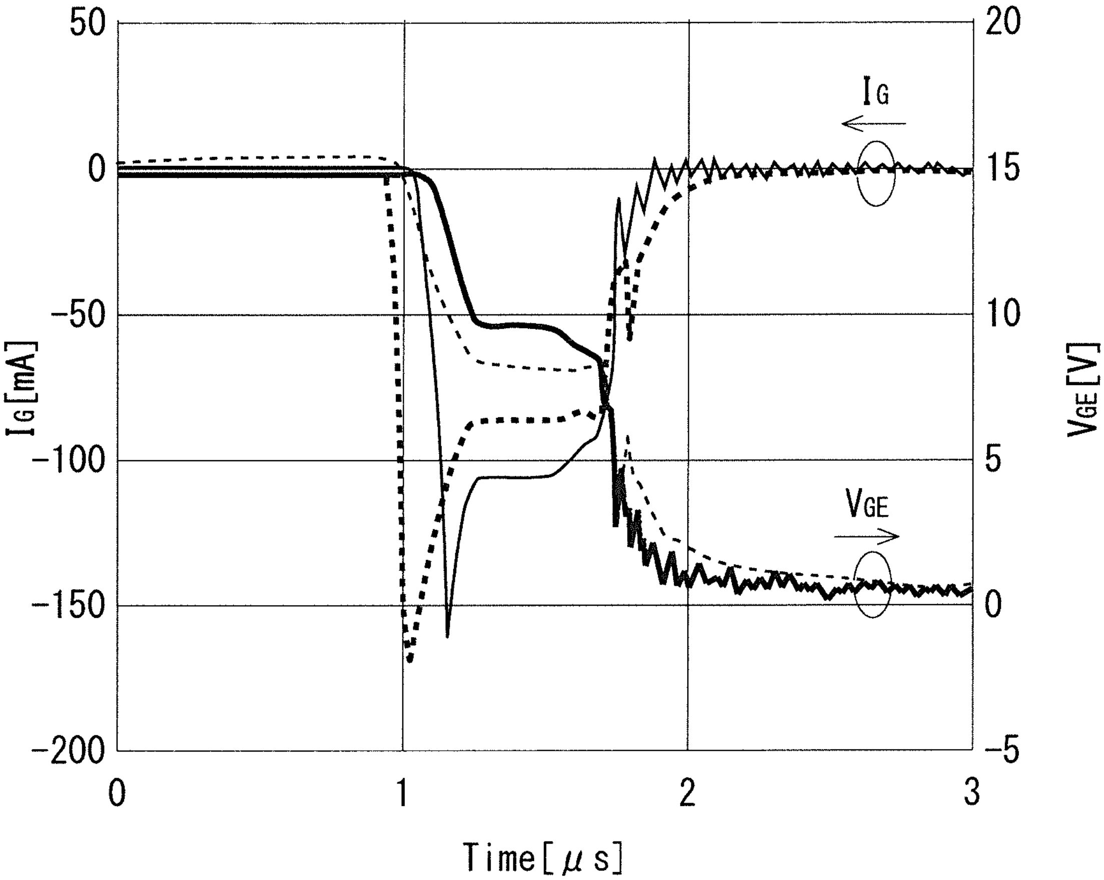


FIG. 14

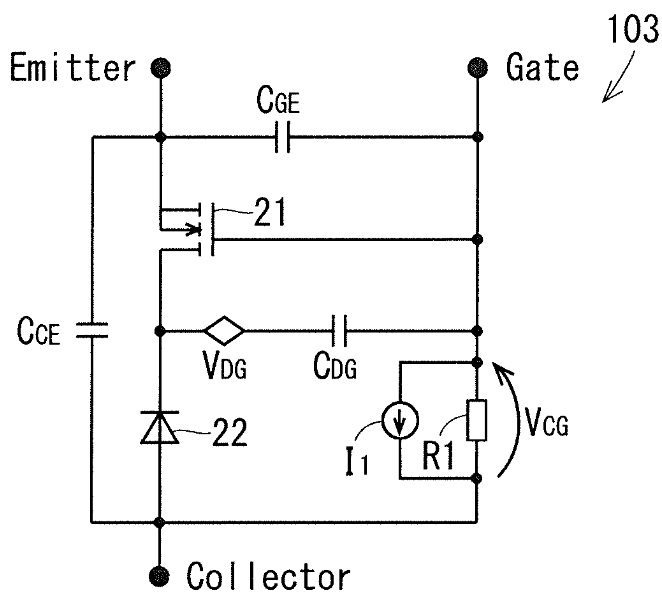


FIG. 15

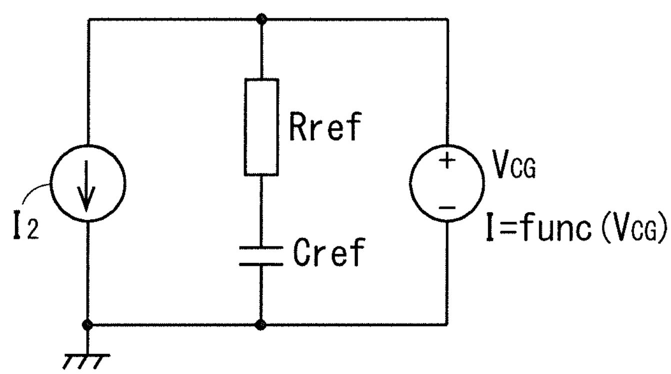


FIG. 16

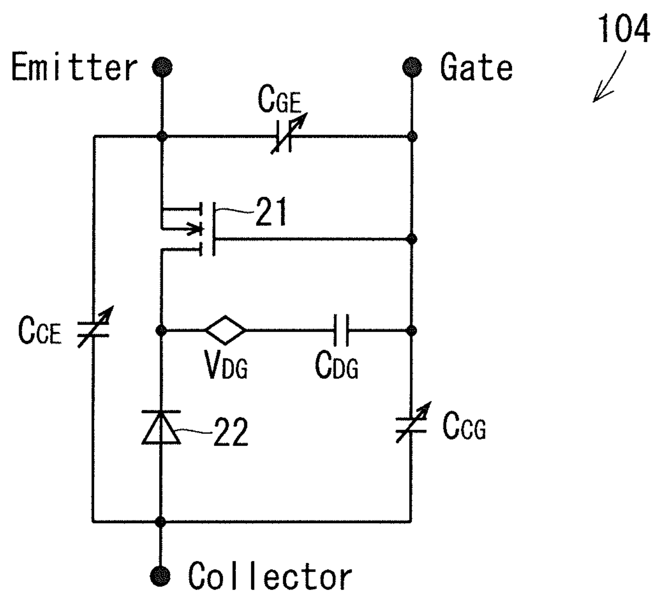
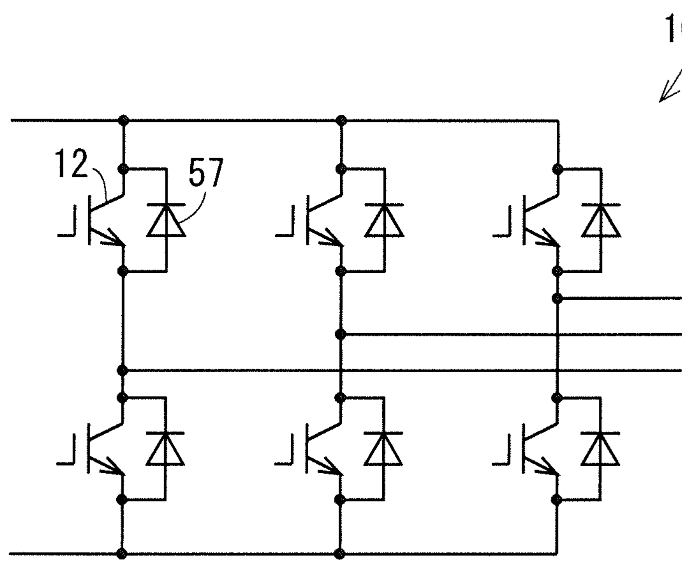


FIG. 17



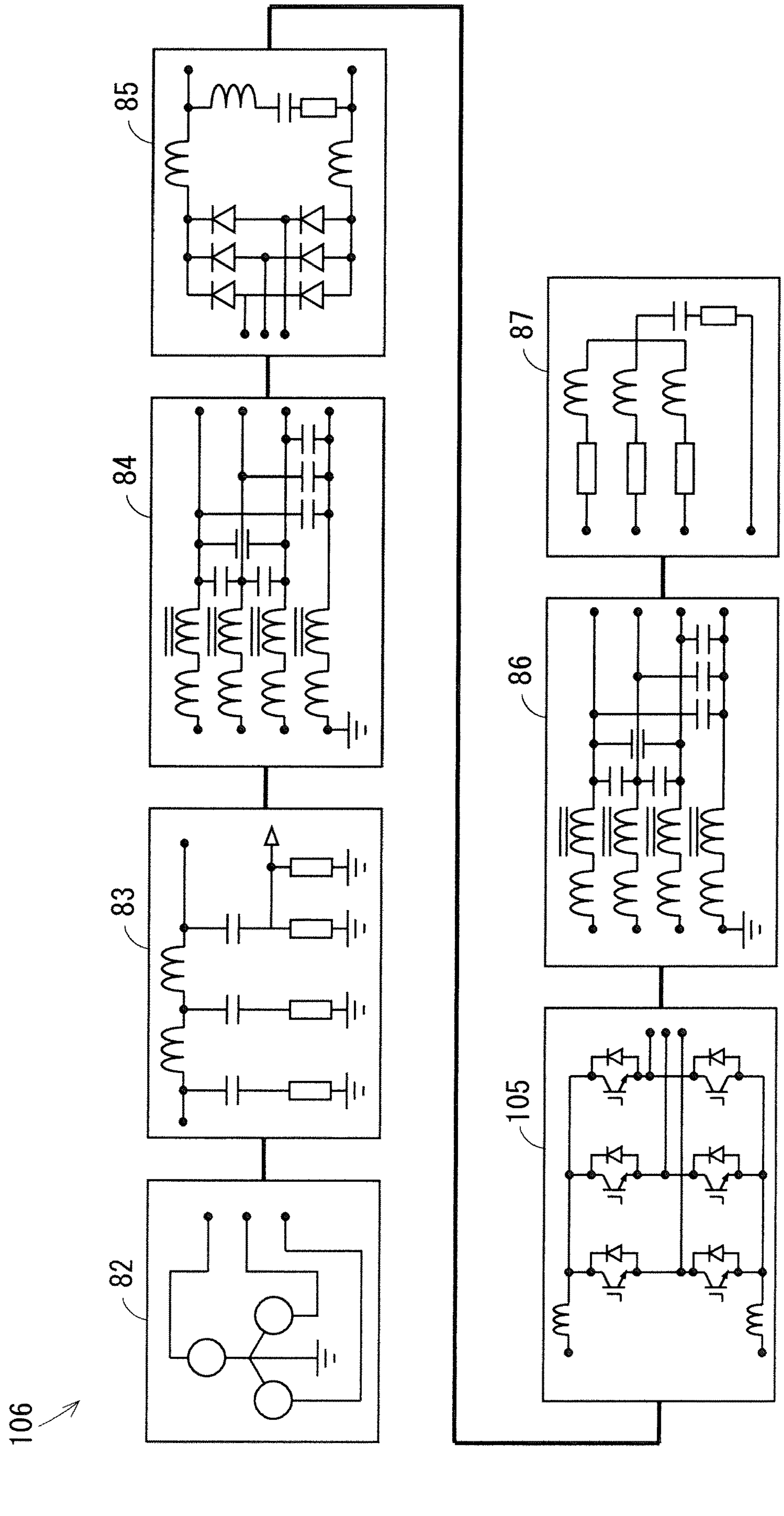
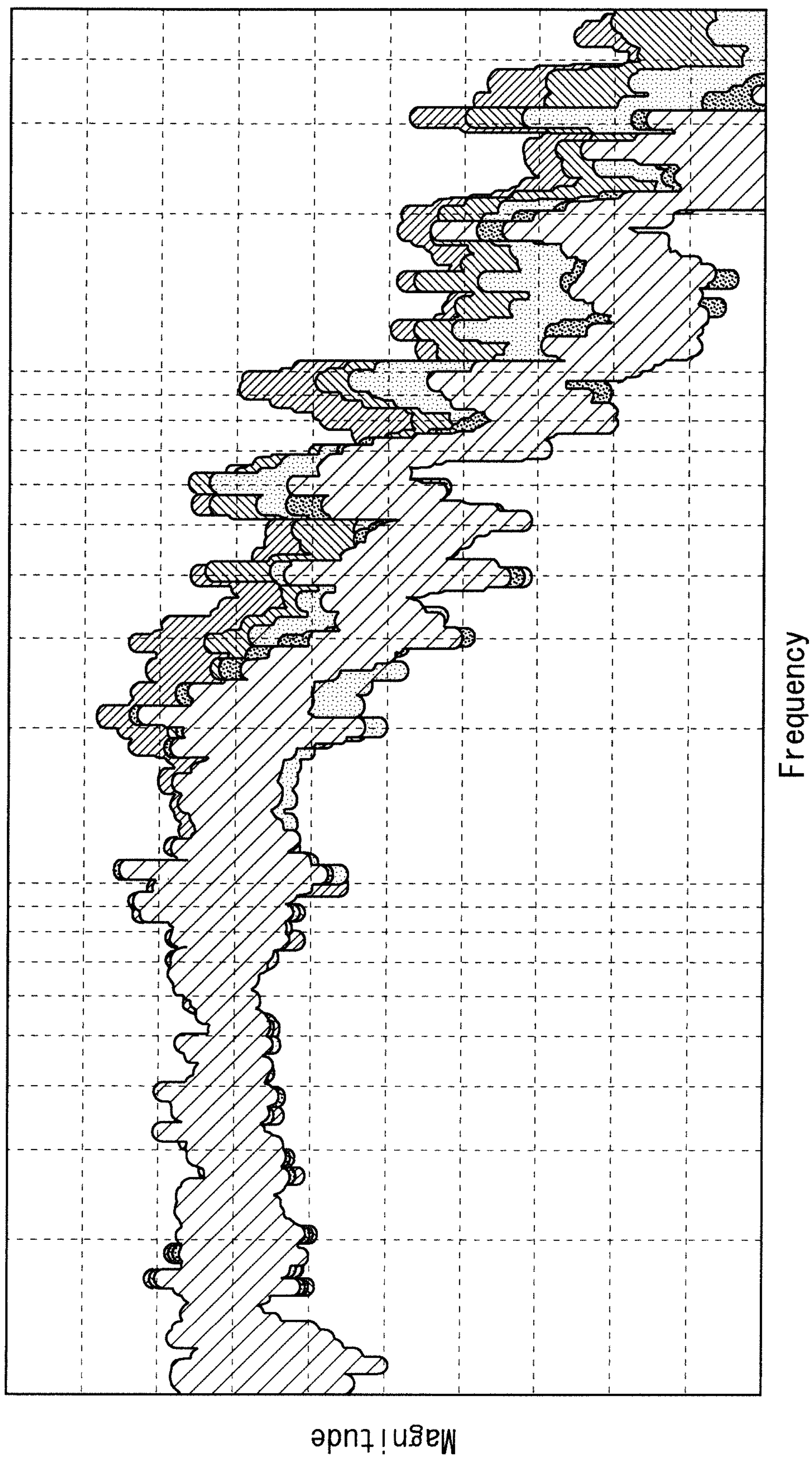


FIG. 18

FIG. 19



SIMULATION MODEL AND SIMULATION METHOD

BACKGROUND OF THE INVENTION

Field of the Invention

The present disclosure relates to a simulation of Carrier Stored Trench Bipolar Transistor (CSTBT).

Description of the Background Art

Generally, in the development of a power electronics device such as an inverter, a circuit configuration is first subject to simulation analysis and then to verification by trial production evaluation.

For the above simulation analysis, for example, a circuit simulation using a Simulation Program with Integrated Circuit Emphasis (SPICE) model is used.

The SPICE model is a model in which the electrical characteristics of a power semiconductor device such as a diode, a metal-oxide-semiconductor field-effect transistor (that is, a MOSFET), an insulated gate bipolar transistor (that is, an IGBT) are simulated and calculated.

For the accurate simulation of the electrical characteristics, the physical parameters of the device model are required to be extracted. Therefore, advanced knowledge of semiconductor physics is required.

However, in general, circuit designers are often not so much required to have knowledge about semiconductor physics, accordingly, a method that makes the extraction of physical parameters with high accuracy possible even without knowledge of semiconductor physics is required. As a method for solving such a problem, for example, a method described in Japanese Patent Application Laid-Open No. 2020-88080 is known.

The behavioral modeling of the IGBT described in Japanese Patent Application Laid-Open No. 2020-88080 does not reflect the Carrier Store (CS) layer; therefore, there is a problem that the operation of the CSTBT having the CS layer cannot be accurately expressed.

SUMMARY

The object of the technique of the present disclosure is to accurately simulate the operation of a CSTBT.

A simulation model of the present disclosure is a simulation model for simulation evaluating characteristics of a CSTBT being a trench gate type IGBT having a carrier storage layer. The simulation model of the present disclosure includes a MOSFET, a diode, capacitance C_{GE}, capacitance C_{CG}, capacitance C_{CE}, capacitance C_{DG}, and a behavioral power supply V_{DG}. The cathode of the diode is connected to the drain of the MOSFET. The capacitance C_{GE} is connected between the source and the gate of the MOSFET and representing gate-emitter capacitance of the CSTBT. The capacitance C_{CG} is connected between the gate of the MOSFET and the anode of the diode and representing gate-collector capacitance of the CSTBT. The capacitance C_{CE} is connected between the source of the MOSFET and the anode of the diode and representing collector-emitter capacitance of the CSTBT. The capacitance C_{DG} is connected between the drain and the gate of the MOSFET and representing drain-gate capacitance of the CSTBT. The behavioral power source V_{DG} is connected in series to the capacitance C_{DG} between the drain and the gate of the MOSFET and representing a drain-gate voltage of the

CSTBT. The behavioral power source V_{DG} performs a switching operation when gate-emitter voltage V_{GE} of the CSTBT reaches a predetermined threshold value.

According to the simulation model of the present disclosure, the behavior of the CSTBT is simulated with high accuracy, in which the gate-emitter voltage V_{GE} of the CSTBT sharply increases due to the switching of the behavioral power supply V_{DG}, causing the MOSFET channel to expand immediately and the current to start flowing sharply.

These and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a cross-sectional view illustrating a vertical structure of a CSTBT;

FIG. **2** is a diagram illustrating a simulation circuit of the CSTBT according to Embodiment 1;

FIG. **3** is a diagram illustrating a simulation circuit of a drive circuit of the CSTBT;

FIG. **4** is a diagram illustrating a simulation circuit of a test circuit of the CSTBT at a high frequency;

FIG. **5** is a graph illustrating the actual waveforms of V_{CE} and I_C in the turn-on operation of the CSTBT;

FIG. **6** is a graph illustrating the actual waveforms of V_{GE} and I_G in the turn-on operation of the CSTBT;

FIG. **7** is a graph illustrating the simulation waveforms of V_{CE} and I_C in the turn-on operation of the CSTBT using the simulation circuit of FIG. **2**;

FIG. **8** is a graph illustrating the simulation waveforms of V_{GE} and I_G in the turn-on operation of the CSTBT using the simulation circuit of FIG. **2**;

FIG. **9** is a diagram illustrating a simulation circuit of a CSTBT according to Embodiment 2;

FIG. **10** is a graph illustrating the V_{CG} dependence of C_{CG};

FIG. **11** is a graph illustrating C_{CG} fitting;

FIG. **12** is a graph illustrating the simulation waveforms of V_{CE} and I_C in the turn-on operation of the CSTBT using the simulation circuit of FIG. **9**;

FIG. **13** is a graph illustrating the simulation waveforms of V_{GE} and I_G in the turn-on operation of the CSTBT using the simulation circuit of FIG. **9**;

FIG. **14** is a diagram illustrating a simulation circuit of a CSTBT according to Embodiment 3;

FIG. **15** is a diagram illustrating a circuit for calculating the current of a behavioral current source in the simulation circuit of FIG. **14**;

FIG. **16** is a diagram illustrating a simulation circuit of a CSTBT according to Embodiment 4;

FIG. **17** is a diagram illustrating a simulation circuit of a 6in1 module having the CSTBT;

FIG. **18** is a diagram illustrating a simulation circuit of a conduction noise evaluation system using the simulation circuit of the 6in1 module illustrating in FIG. **17**; and

FIG. **19** is a diagram illustrating an analysis result of conduction noise by the simulation circuit of FIG. **18**.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

A. Embodiment 1

<A-1. Configuration>

FIG. **1** is a cross-sectional view illustrating a vertical structure of a CSTBT **12**. As illustrated in FIG. **1**, the CSTBT **12** includes an emitter electrode **1**, a collector electrode **2**, a gate electrode **3**, a P+ layer **4**, an N+ emitter layer **5**, a channel-doped layer **6**, a carrier store layer (CS layer) **7**, an N− drift layer. **8**, an N+ buffer layer **9**, a P+ collector layer **10**, and a gate oxide film **11**. The CS layer **7**, the channel-doped layer **6**, the N+ emitter layer **5**, and the P+ layer **4** are laminated in this order on a first main surface of the N− drift layer **8**. Trenches are formed that extend through the N+ emitter layer **5**, the channel-doped layer **6**, and the CS layer **7** and reaches the N− drift layer **8**, and in the trenches, the gate electrodes **3** and the gate oxide films **11** formed to cover the gate electrodes **3** are formed. The emitter electrode **1** is formed on the P+ layer **4** and the N+ emitter layer **5**. The N+ buffer layer **9**, the P+ collector layer **10**, and the collector electrode **2** are formed in this order on a second main surface opposite to the first main surface of the N− drift layer **8**.

FIG. **2** illustrates a simulation model **101** of the CSTBT **12** of Embodiment 1. The simulation model **101** is an equivalent circuit of the CSTBT **12** and is used for the simulation of the CSTBT **12**. The simulation model is, for example, input into a computer and further displayed in a simulator.

Main elements of the simulation model **101** of the CSTBT **12** are a MOSFET **21** and a diode **22**. The MOSFET **21** is composed of the N+ emitter layer **5**, the channel-doped layer **6**, the CS layer **7**, the gate oxide film **11**, and the gate electrode **3** of the CSTBT **12**. The diode **22** is composed of the N− drift layer **8**, the N+ buffer layer **9**, and the P+ collector layer **10** of the CSTBT **12**.

The gate of MOSFET **21** corresponds to the gate of the CSTBT **12**, and the emitter of MOSFET **21** corresponds to the emitter of the CSTBT **12**. The cathode of the diode **22** is connected to the drain of the MOSFET **21**. The anode of diode **22** corresponds to the collector of the CSTBT **12**.

The gate-emitter capacitance C_{GE} of the CSTBT **12** is connected between the gate and emitter of the MOSFET **21**.

The behavioral power source V_{DG} and the drain-gate capacitance C_{DG} of the CSTBT **12** are connected in series between the drain and gate of MOSFET **21**. The behavioral power source V_{DG} is a behavioral power source having a function of switching with an arbitrary voltage as a threshold value. For example, when the gate-emitter voltage V_{GE} reaches the threshold voltage V_{th} of the CSTBT **12**, the behavioral power source V_{DG} performs switching. Or, when the gate-emitter voltage V_{GE} falls below the threshold voltage V_{th} of the CSTBT **12**, the behavioral power source V_{DG} performs switching. It should be noted that, the behavioral power source V_{DG} does not have to perform switching. The behavioral power source V_{DG} can take a positive, negative, or 0 value with respect to the gate potential.

The drain-gate capacitance C_{DG} is formed by CS layer **7** of the CSTBT **12**. Although any value may be entered for the drain-gate capacitance C_{DG}, the design value is applied, for example.

The gate-collector capacitance C_{CG} of the CSTBT **12** is connected between the gate of the MOSFET **21** and the anode of the diode **22**.

The collector-emitter capacitance C_{CE} of the CSTBT **12** is connected between the source of the MOSFET **21** and the anode of the diode **22**.

FIG. **3** is a diagram illustrating a simulation model of a drive circuit **30** of the CSTBT **12**. The drive circuit **30** is a gate drive circuit that applies a gate voltage to the gate terminal of the CSTBT **12**, in other words, the MOSFET **21**. The simulation model of the drive circuit **30** illustrated in FIG. **3** is an equivalent circuit of the drive circuit **30**, and the configuration thereof includes an optocoupler **31**, PNP transistors **32***a*, **32***b*, NPN transistors **33***a*, **33***b*, a MOSFET **34**, diodes **35***a*, **35***b*, resistors **36***a*, **36***b*, **36***c*, and a power source **37**.

The optocoupler **31** receives an input signal. The output of the optocoupler **31** is input to the bases of the PNP transistor **32***a* and the NPN transistor **33***a*. The emitter of the PNP transistor **32***a* and the emitter of the NPN transistor **33***a* are connected to the gate of the MOSFET **34**. The collector of the NPN transistor **33***a* is connected to the power supply potential of the power source **37**. The drain of the MOSFET **34** is connected to the power supply potential via the resistor **36***c* and is connected to the bases of the PNP transistor **32***b* and the NPN transistor **33***b*. The source of MOSFET **34** is connected to the reference potential.

The collector of the NPN transistor **33***b* is connected to the power supply potential, and the collector of the PNP transistor **32***b* is connected to the reference potential. The emitters of the PNP transistor **32***b* and the NPN transistor **33***b* are connected to the anode of the diode **35***a* and the cathode of the diode **35***b*, respectively. The cathode of the diode **35***a* is connected to the resistor **36***a*. The anode of the diode **35***a* is connected to the resistor **36***b*. Of both sides of the resistors **36***a* and **36***b*, the side opposite the side on which the diodes **35***a* and **35***b* are respectively connected is connected to the output terminal.

FIG. **4** is a diagram illustrating a simulation model **80** of a test circuit of the CSTBT **12**. The simulation model **80** is an equivalent circuit of the test circuit of the CSTBT **12** at a high frequency. The simulation model **80** includes a substrate **50** to which the CSTBT **12** and the freewheeling diode **57** are die-bonded, the drive circuit **30**, an inductive load **60**, and a power supply circuit **70**. The substrate **50** is represented by the freewheeling diode **57**, an inductance **51** and an inductance **52** connected to the cathode and anode of the freewheeling diode **57**, respectively, an inductance **53** connected to the inductance **52**, the CSTBT **12** whose collector is connected to the inductance **53**, and an inductance **55** and an inductance **54** connected to the gate and emitter of CSTBT **12**, respectively. The inductance **54** is connected between the emitter of CSTBT **12** and the reference potential.

One output terminal of the drive circuit **30** is connected to the inductance **55** via an inductance **41**, and the other output terminal is connected to the reference potential via an inductance **42**. In FIG. **4**, the simulation model **101** illustrated in FIG. **2** is applied to CSTBT **12**.

The inductance **51** is connected between a first terminal T**1** and the freewheeling diode **57** of the substrate **50**. The inductance **52** is connected between a second terminal T**2** and the freewheeling diode **57** of the substrate **50**. A third terminal T**3** of the substrate **50** is connected to the reference potential.

The inductive load **60** is connected between the first terminal T**1** and the second terminal T**2** of the substrate **50**, and the power supply circuit **70** is connected between the first terminal T**1** and the third terminal T**3**. The inductive load **60** is represented by a parallel connection of a series connection of a resistor **64**, an inductance **62** and a capacitance **66**, an inductance **61**, a resistor **63** and a capacitance **65**. The power supply circuit **70** is represented by an inductance **71**, an electric field capacitor **72** and a resistor **73**.

<A-2. Operation>

Representative examples of double pulse test results are illustrated in FIGS. **5** and **6**. FIG. **5** illustrates measured values of the V_{CE} and I_C in the CSTBT **12**. FIG. **6** illustrates measured values of the V_{GE} and I_G in the CSTBT **12**. In FIGS. **5** and **6**, the horizontal axis represents time [μs]. In FIG. **5**, the vertical axes represent I_C [A] and V_{CE} [V], and in FIG. **6**, the vertical axes represent I_G [A] and V_{GE} [V]. As illustrated in FIG. **5**, I_G of 15 [A] is flowing when V_{CE} of 300 [V] is applied. As illustrated in FIG. **6**, when the V_{GE} reaches an arbitrary voltage, the V_{GE} sharply increases due to the switching of the V_{DG}, so that the MOSFET channel immediately expands and the current starts to flow sharply.

FIGS. **7** and **8** illustrate the simulation results of the double pulse test using the simulation model **80** of FIG. **4**. The horizontal and vertical axes of FIG. **7** are the same as the horizontal and vertical axes of FIG. **5**, and the horizontal and vertical axes of FIG. **8** are the same as the horizontal and vertical axes of FIG. **6**. In FIGS. **7** and **8**, the broken lines indicate the simulation result, and the solid lines indicate the measured values illustrated in FIGS. **5** and **6**. From FIGS. **7** and **8**, it can be seen that the simulation model **101** of FIG. **2** simulates the behaviors of the V_{CE}, I_C, V_{GE}, and I_G in the CSTBT **12** with high accuracy.

<A-3. Effect>

The simulation model **101** of the CSTBT **12** of Embodiment 1 includes the MOSFET **21**, the diode **22** whose cathode is connected to the drain of the MOSFET **21**, the capacitance C_{GE} connected between the source and gate of the MOSFET **21** and representing the gate-emitter capacitance of the CSTBT **12**, the capacitance C_{CG} connected between the gate of the MOSFET **21** and the anode of the diode **22** and representing the gate-collector capacitance of the CSTBT **12**, the capacitance C_{CE} connected between the source of the MOSFET **21** and the anode of the diode **22** and representing the collector-emitter capacitance of the CSTBT **12**, the capacitance C_{DG} connected between the drain and gate of the MOSFET **21** and representing the drain-gate capacitance of the CSTBT **12**, and the behavioral power source V_{DG} connected in series to the capacitance C_{DG} between the drain and gate of the MOSFET **21** and representing the drain-gate voltage of the CSTBT **12**.

Then, the behavioral power source V_{DG} performs a switching operation when the gate-emitter voltage V_{GE} of the CSTBT **12** reaches a predetermined threshold value. Therefore, according to the simulation model **101**, the behavior of the CSTBT **12** is simulated with high accuracy, in which the gate-emitter voltage V_{GE} of the CSTBT **12** sharply increases due to the switching of the behavioral power supply V_{DG}, causing the MOSFET channel to expand immediately and the current to start flowing sharply.

Also, the simulation model **101** may include a gate drive circuit that applies a voltage to the gate of the MOSFET **21**. This allows simulating the gate voltage and the gate current of the CSTBT **12** with high accuracy.

B. Embodiment 2

<B-1. Configuration>

FIG. **9** is a diagram illustrating a simulation model **102** of the CSTBT **12** of Embodiment 2. In the simulation model **102** of the CSTBT **12**, a variable capacitance that changes depending on the gate-collector voltage V_{CG}, is applied to the gate-collector capacitance C_{CG} of the CSTBT **12** in the simulation model **101** described in Embodiment 1, and except for that regard, the configuration is the same as the simulation model **101**.

FIG. **10** illustrates the actual measurement results of the gate-collector capacitance C_{CG} of the CSTBT **12**. The horizontal axis of FIG. **10** represents the gate-collector voltage V_{CG} [V], and the vertical axis represents the gate-collector capacitance C_{CG} [F]. As illustrated in FIG. **10**, the gate-collector capacitance C_{CG} varies depending on the gate-collector voltage V_{CG}. In order to reflect this phenomenon, the simulation model **102** in FIG. **9** represents the gate-collector capacitance C_{CG} by the following formula with the gate-collector voltage V_{CG} as a variable. Note that Ca, Ct, Vt, and Vc are arbitrary fixed values, y=arc tan (x) is the inverse function of y=tan (x), and π represents the circular constant.

$$C_{CG}=Ca\cdot(1-Ct*2/\pi)\cdot\arctan\{(V_{CG}-Vt)/Vc\} \quad \text{[Expression 1]}$$

According to Expression 1, the gate-collector capacitance C_{CG} is fitted to a value close to the measured value as illustrated in FIG. **11**.

<B-2. Operation>

FIGS. **12** and **13** illustrate the simulation results using the simulation model **102** of the CSTBT **12**. That is, simulation of the behaviors of V_{CE}, I_C, V_{GE}, and I_G in the CSTBT **12** was performed with the application of the simulation model **102** to the CSTBT **12** in the simulation model **80** of FIG. **4** in the same manner as in Embodiment 1. In FIGS. **12** and **13**, the horizontal axis represents time [μs]. In FIG. **12**, the vertical axes represent I_C [A] and V_{CE} [V], and in FIG. **13**, the vertical axes represent I_G [A] and V_{GE} [V]. In FIGS. **12** and **13**, the broken lines indicate the simulation result, and the solid lines indicate the measured values. From FIGS. **12** and **13**, it can be seen that the simulation model **102** simulates the behaviors of the V_{CE}, I_C, V_{GE}, and I_G in the CSTBT **12** with high accuracy.

<B-3. Effect>

In the simulation model **102** of the CSTBT **12** of Embodiment 2, the gate-collector capacitance C_{CG} of the CSTBT **12** changes depending on the gate-collector voltage V_{CG} of the CSTBT **12**. Therefore, according to the simulation model **102**, the gate-collector capacitance C_{CG} that changes depending on the gate-collector voltage V_{CG} of the CSTBT **12** can be accurately reflected.

Further, in the simulation model **102**, the gate-collector capacitance C_{CG} of CSTBT **12** is represented by Ca(1−Ct*2/π) arc tan {(V_{CG}−Vt)/Vc}, with Ca, Ct, Vt, and Vc as constants and the gate-collector voltage V_{CG} of the CSTBT **12** as variables. In this manner, according to the simulation model **102**, the gate-collector capacitance C_{CG} is represented by a continuous function of the gate-collector voltage V_{CG}, so the behaviors of V_{CE}, I_C, V_{GE}, and I_G can be calculated with high accuracy and stability.

C. Embodiment 3

<C-1. Configuration>

FIG. **14** is a diagram illustrating a simulation model **103** of the CSTBT **12** of Embodiment 3. In the simulation model **103** of the CSTBT **12**, the gate-collector capacitance C_{CG} of the CSTBT **12** is represented by a parallel connection of the behavioral current source I_1 and a resistor R**1** based on the gate-collector voltage V_{CG}, in the simulation model **101** described in Embodiment 1, and except for that regard, the configuration is the same as the simulation model **101**. The behavioral current source I_1 is also referred to as a first behavioral current source.

The current of the behavioral current source I_1 in the simulation model **103** is calculated using the circuit illustrated in FIG. **15**. The circuit illustrated in FIG. **15** is composed of a parallel connection of a series connection of a reference resistor Rref and a reference capacitance Cref, the behavioral voltage source V_{CG} and a behavioral current source I_2. The behavioral current source I_2 is also referred to as a second behavioral current source. The current of the behavioral current source I_2 is represented by the gate-collector capacitance C_{CG}, the reference capacitance Cref, and the time derivative of the gate-collector voltage V_{CG}. The current I=func (V_{CG}) flowing through the behavioral voltage source V_{CG} of the circuit illustrated in FIG. **15** corresponds to the current of the behavioral current source I_1 in the simulation model **103** of the CSTBT **12**.

<C-2. Effect>

In the simulation model **103** of the CSTBT **12** of Embodiment 3, the gate-collector capacitance C_{CG} of the CSTBT **12** is represented by a parallel connection of the behavioral current source I_1 being the first behavioral current source and the resistor R**1**, and, in the circuit composed of the series connection of the reference resistor Rref and the reference capacitance Cref, the behavioral current source I_2 being the second behavioral current source connected to both ends of the series connection, and the behavioral voltage source V_{CG} representing the gate-collector voltage of the CSTBT **12** connected to both ends of the series connection, the current flowing through the second behavioral current source I_2 is represented by the gate-collector capacitance C_{CG}, the reference capacitance Cref, and the time derivative of the gate-collector voltage V_{CG} of the CSTBT **12**, and the current flowing through the behavioral voltage source V_{CG} corresponds to the current of the behavioral current source I_1.

According to the simulation model **103**, the gate-collector capacitance C_{CG} can be calculated as a function of the gate-collector voltage V_{CG} as a variable with the gate-collector capacitance C_{CG} of the CSTBT **12** as a voltage variable capacitor. In addition, the gate-collector voltage V_{CG} is applied to the reference resistor Rref and the reference capacitance Cref instead of the behavioral current source I_2; therefore, the calculation can be stabilized.

D. Embodiment 4

<D-1. Configuration>

FIG. **16** is a diagram illustrating a simulation model **104** of the CSTBT **12** of Embodiment 4. The simulation model **104** of the CSTBT **12** is represented by the collector-emitter capacitance C_{CE} and the gate-emitter capacitance C_{GE} as variable capacitances with the voltage applied to each component as variables in the simulation model **102** described in Embodiment 2.

The collector-emitter capacitance C_{CE} is represented by the following expression with Ca, Ct, Vt, and Vc as arbitrary fixed values and V_{CE} as a variable.

$$C_{CE}=Ca\cdot(1-Ct*2/\pi)\cdot\arctan\{(V_{CE}-Vt)/Vc\} \quad \text{[Expression 2]}$$

The gate-emitter capacitance C_{GE} is represented by the following expression with Ca, Ct, Vt, and Vc as arbitrary fixed values and V_{GE} as a variable.

$$C_{GE}=Ca\cdot(1-Ct*2/\pi)\cdot\arctan\{(V_{GE}-Vt)/Vc\} \quad \text{[Expression 3]}$$

In FIG. **16**, although both the collector-emitter capacitance C_{CE} and the gate-emitter capacitance C_{GE} are represented as variable capacitances, only one of them may be represented as a variable capacitance.

<D-2. Effect>

In the simulation model **104** of the CSTBT **12** of Embodiment 4, the collector-emitter capacitance C_{CE} of the CSTBT **12** may change depending on the collector-emitter voltage V_{CE} of the CSTBT **12**. Further, the gate-emitter capacitance C_{GE} of the CSTBT **12** may change depending on the gate-emitter voltage V_{GE} of the CSTBT **12**. With such a configuration, according to the simulation model **104** of the CSTBT **12**, the stable representation of the V_{GE} dependence of C_{GE} or the V_{CE} dependence of C_{CE} is ensured.

E. Embodiment 5

<E-1. Configuration>

FIG. **17** illustrates a simulation model **105** of a semiconductor module of Embodiment 5. The semiconductor module represented by the simulation model **105** is a 6in1 module to which the CSTBT **12** is applied. That is, the 6in1 module has six pairs consisting of the CSTBT **12** and the freewheeling diode **57** connected in antiparallel to the CSTBT **12**. To the CSTBT **12** in the simulation model **105**, the simulation model **101** to **104** of the CSTBT **12** in any one of Embodiments 1 to 4 is applied.

<E-2. Effect>

The simulation model **105** of the semiconductor module of Embodiment 5 is a simulation model of a 6in1 having the CSTBT **12**, and any simulation model **101** to **104** of any of Embodiment 1 to 4 is applied to the CSTBT **12** of the 6in1 module. Therefore, according to the simulation model **105**, the behavior of the CSTBT **12** can be represented as the 6in1 module.

F. Embodiment 6

<F-1. Configuration>

FIG. **18** illustrates a simulation model **106** of the conduction noise evaluation system of Embodiment 6. As illustrated in FIG. **18**, the simulation model **106** includes a power supply circuit model **82**, an LISN model **83**, a first cable model **84**, a rectifier model **85**, a 6in1 module simulation model **105**, a second cable model **86**, and a motor model **87**.

The power supply circuit model **82** is, for example, a simulation model of a three-phase power supply circuit. The LISN model **83** is a simulation model of the LISN, and is provided in the following stage of the power supply circuit model **82**. The first cable model **84** is, for example, a simulation model of a three-phase four-wire cable, and is provided in the following stage of the LISN model **83**. The rectifier model **85** is a simulation model of a rectifier and a smoothing capacitor, and is provided in the following stage of the first cable model **84**. The simulation model **105** is a simulation model of the 6in1 module described in Embodiment 5, and is provided in the following stage of the rectifier model **85**. The second cable model **86** is, for example, a simulation model of a three-phase four-wire cable, and is provided in the following stage of the simulation model **105**. The motor model **87** is a simulation model of a motor, and is provided in the following stage of the second cable model **86**.

Conductive noise can be detected at the output terminal of the LISN by performing analysis with Transient using the simulation model **106** illustrated in FIG. **18**. In addition, the profile of conduction noise is output by performing frequency transformation such as Discrete Fourier Transform

(DFT), Fast Fourier Transform (FFT), or wavelet transformation on at least a part of the detected conduction noise.

<F-2. Operation>

According to the simulation model **106**, analysis of conduction noise (noise terminal voltage) or common mode current in accordance with the characteristics of the CSTBT **12** incorporated in a Dual-In-Line Package Intelligent Power Module (DIPIPM) such as 6in1 module can be performed. Then, according to the simulation model **106**, identification of the dominant part of the frequency domain determined as noise can be performed by analyzing the current in the common mode or the differential mode using the power device as the signal source.

FIG. **19** illustrates the results of simulating the conduction noise in the simulation model **106** for a plurality of cases in which the concentrations of the CS layer **7** of the CSTBT **12** are different.

<F-3. Effect>

The simulation model **106** of the conduction noise evaluation system of Embodiment 6 includes the power supply circuit model **82** being a simulation model of a power supply circuit, the LISN model **83** provided in the following stage of the power supply circuit model **82** and being a simulation model of an LISN, the first cable model **84** provided in the following stage of the LISN model **83** and being a simulation model of a cable, the rectifier model **85** provided in the following stage of the first cable model **84** and being a simulation model of a rectifier and a smoothing capacitor, the simulation model **105** provided in the following stage of the rectifier model **85**, the second cable model **86** provided in the following stage of the simulation model **105** and being a simulation model of a cable, and the motor model **87** being a simulation model of a motor, and provided in the following stage of the second cable model **86**. Therefore, according to the simulation model **106**, evaluation of the conduction noise in accordance with the characteristics of the CSTBT **12** can be performed.

The embodiments can be combined, appropriately modified or omitted, without departing from the scope of the invention.

While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

What is claimed is:

1. A simulation model for simulation evaluating characteristics of a CSTBT being a trench gate type IGBT having a carrier storage layer, the simulation model comprising:
- a MOSFET:
- a diode whose cathode is connected to a drain of the MOSFET;
- capacitance C_{GE} connected between a source and a gate of the MOSFET and representing gate-emitter capacitance of the CSTBT;
- capacitance C_{CG} connected between a gate of the MOSFET and an anode of the diode and representing gate-collector capacitance of the CSTBT;
- capacitance C_{CE} connected between a source of the MOSFET and the anode of the diode and representing collector-emitter capacitance of the CSTBT;
- capacitance C_{DG} connected between the drain and the gate of the MOSFET and representing drain-gate capacitance of the CSTBT; and
- a behavioral power source V_{DG} connected in series to the capacitance C_{DG} between the drain and the gate of the MOSFET and representing drain-gate voltage of the CSTBT, wherein
- the behavioral power source V_{DG} performs a switching operation when a voltage is applied to the gate of the MOSFET and gate-emitter voltage V_{GE} of the CSTBT reaches a predetermined threshold value.

2. The simulation model according to claim **1**, wherein
- the capacitance C_{CG} varies depending on the gate-collector voltage V_{CG} of the CSTBT.

3. The simulation model according to claim **2**, wherein
- the capacitance CG is represented by $Ca(1-Ct*2/\pi)$ arc tan $\{(V_{CG}-Vt)/Vc\}$, with Ca, Ct, Vt, and Vc as constants and the gate-collector voltage V_{CG} of the CSTBT as variables.

4. The simulation model according to claim **1**, wherein
- the capacitance C_{CG} is represented by a parallel connection of a first behavioral current source and a resistor, and,
- in a circuit composed of a series connection of a reference resistor and a reference capacitance,
- a second behavioral current source connected to both ends of the series connection, and
- a behavioral voltage source representing the gate-collector voltage V_{CG} of the CSTBT connected to both ends of the series connection,
- a current flowing through the second behavioral current source is represented by time derivative of the capacitance C_{CG}, the reference capacitance, and the gate-collector voltage V_{CG} of the CSTBT, and
- a current flowing through the behavioral voltage source corresponds to a current of the first behavioral current source.

5. The simulation model according to claim **1**, wherein
- the capacitance C_{CE} varies depending on a collector-emitter voltage V_{CE} of the CSTBT.

6. The simulation model according to claim **1**, wherein
- the capacitance C_{GE} varies depending on the gate-emitter voltage V_{GE} of the CSTBT.

7. A simulation model of a 6in1 module having the CSTBT, wherein
- the simulation model according to claim **1** is applied to the CSTBT of the 6in1 module.

8. The simulation model according to claim **1**, further comprising
- a gate drive circuit that applies a voltage to the gate of the MOSFET.

9. The simulation model comprising:
- a power supply circuit model being a simulation model of a power supply circuit;
- an LISN model provided in a following stage of the power supply circuit model and being a simulation model of an LISN;
- a first cable model provided in a following stage of the LISN model and being a simulation model of a cable;
- a rectifier model provided in a following stage of the first cable model and being a simulation model of a rectifier and a smoothing capacitor;
- a simulation model according to claim **7**, provided in a following stage of the rectifier model;
- a second cable model provided in a following stage of the simulation model according to claim **7** and being a simulation model of a cable; and
- a motor model provided in a following stage of the second cable model and being a simulation model of a motor.

10. A simulation method for evaluating characteristics of a CSTBT being a trench gate type IGBT having a carrier storage layer, wherein the simulation method evaluates characteristics of the CSTBT using a simulation model, the simulation model including:

a MOSFET, a diode whose cathode is connected to a drain of the MOSFET, capacitance C_{GE} connected between a source and a gate of the MOSFET and representing gate-emitter capacitance of the CSTBT, capacitance C_{CG} connected between a gate of the MOSFET and an anode of the diode and representing gate-collector capacitance of the CSTBT, capacitance C_{CE} connected between a source of the MOSFET and the anode of the diode and representing collector-emitter capacitance of the CSTBT, capacitance C_{DG} connected between the drain and the gate of the MOSFET and representing drain-gate capacitance of the CSTBT, and a behavioral power source V_{DG} connected in series to the capacitance C_{DG} between the drain and the gate of the MOSFET and representing drain-gate voltage of the CSTBT, and

the method comprising:

applying a voltage to the gate of the MOSFET; and

operating the behavioral power source V_{DG} to perform a switching operation when gate-emitter voltage V_{GE} of the CSTBT reaches a predetermined threshold value.

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